Indian Institute of Information Technology Vadodara
Mohd Anas

Education

Indian Institute of Information Technology Vadodara — **Expected Graduation: Aug 2027**
B.Tech, Information Technology ***7.46 / 10.0 GPA*** — *Gujarat, India*

- Coursework: Backend System, Distributed System, Operating Systems, Object-Oriented Programming, Data Structures & Algorithms, Embedded System, Machine Learning

Experience

Aarogya Kaya — **July 2025 – Jan 2026**
SDE Intern — *Remote*

- Architected Go Backend services for Shiprocket and Instagram Graph API integrations, implementing CQRS architecture and a three-tier caching strategy (in-memory, Redis, and database), reducing average read latency from 420ms to 38ms (91%) and maintaining p95 response times below 250ms.
- Refactored 15+ Backend modules using SOLID principles and Design Patterns (such as Dependency Injection), reducing code duplication by 30% and improving feature delivery speed.
- Engineered fault-tolerant asynchronous synchronization pipelines with retry mechanisms and idempotent processing, supporting 500+ daily events while maintaining 99.9% data consistency.

Open Source Contributor — **July 2025 – Present**
Contributor — *Remote*

- Contributed a merged Pull Request to Next.js (Vercel), migrating the patch-next utility from CommonJS to TypeScript ESM and reducing contributor setup overhead by eliminating repeated dependency reinstalls.
- Contributed 22 merged Pull Requests and opened 12 Issues across open-source projects, fixing bugs, implementing features, and improving developer tooling.

Projects

a3.store | *Distributed Key-value Store — Python, EKS, AWS, SQLite, Prometheus* | GitHub | Live

- Developed a leaderless Distributed key-value store implementing replication, failure detection, synchronization, gossip-based membership, and anti-entropy repair across a 3-node cluster.
- Deployed the system on AWS EKS using Kubernetes StatefulSets and persistent volumes, managing rolling upgrades, pod restarts, and self-healing behavior.
- Architected internal modules using SOLID principles and clean interfaces, decoupling storage engines from network transport to enable pluggable consensus mechanisms and isolated unit testing.

Shop-Intelligence | *Customer Behavior Analytics using Computer Vision — Python, Next.js, Pytest* | GitHub | Live

- Built a distributed real-time video-processing pipeline for four synchronized CCTV streams using concurrent workers and message queues, processing 15 FPS while reducing frame-processing latency to 50 ms.
- Designed an event-driven analytics engine and global visitor registry that generated real-time backend events powering customer behavior dashboards and retail insights.
- Implemented concurrent processing with multiprocessing and multi-threading to maximize throughput and ensure scalable multi-camera analytics.

Achievements

- Achieved Codeforces Pupil (1292).
- Solved 400+ DSA problems in C++ across Codeforces, LeetCode, CodeChef, and GeeksforGeeks.

Technical Skills

- **Languages**: Python, Go, C, C++, TypeScript, JavaScript, SQL
- **Backend / Frontend**: Golang, FastAPI, gRPC, REST APIs, Node.js, React
- **Tools & Technologies**: Git, GitHub, Docker, Kubernetes, AWS, EC2, EKS, IAM, VPC, Terraform, Linux, Kafka, Jenkins, Redis, SQLite, Prometheus, Grafana, Loki
- **AI & LLMs**: Antigravity, Claude Code, RAG, Prompt Engineering, Deep Learning, Agentic AI, LangChain, LangGraph, Ollama

Leadership & Activities

Technical Committee, IIIT Vadodara

Core Member

- Organized an intra-college hackathon and mentored 25+ teams as a Technical Committee Core Member.
- Organized 5+ events and workshops, driving student engagement in competitive programming as Coding Club Member.
- Led recruitment, mentorship, and event initiatives for 100+ students as Joint Secretary of DOT Club.